

Fine-tuning leaves a traceable family history in model weights.

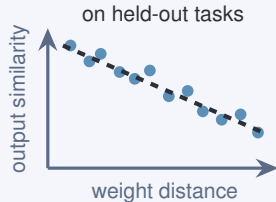
A. Can weights estimate the tree?



✓ every true clade estimated in 85–100% of trees

(3% for a random tree); one weight matrix,
0.1–0.2% of the weights, suffices

B. Do weights track behavior?



✓ closer weights, closer outputs

significant on every held-out
task for Llama; mixed for Flan

C. When does estimation work?



✓ label-free: no true tree needed

estimation odds $\times 5.9$ per
+0.1 in tree-likeness A